Credit Card Behaviour Score Prediction: Comprehensive Project Report

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1. Process Overview & Modeling Strategy

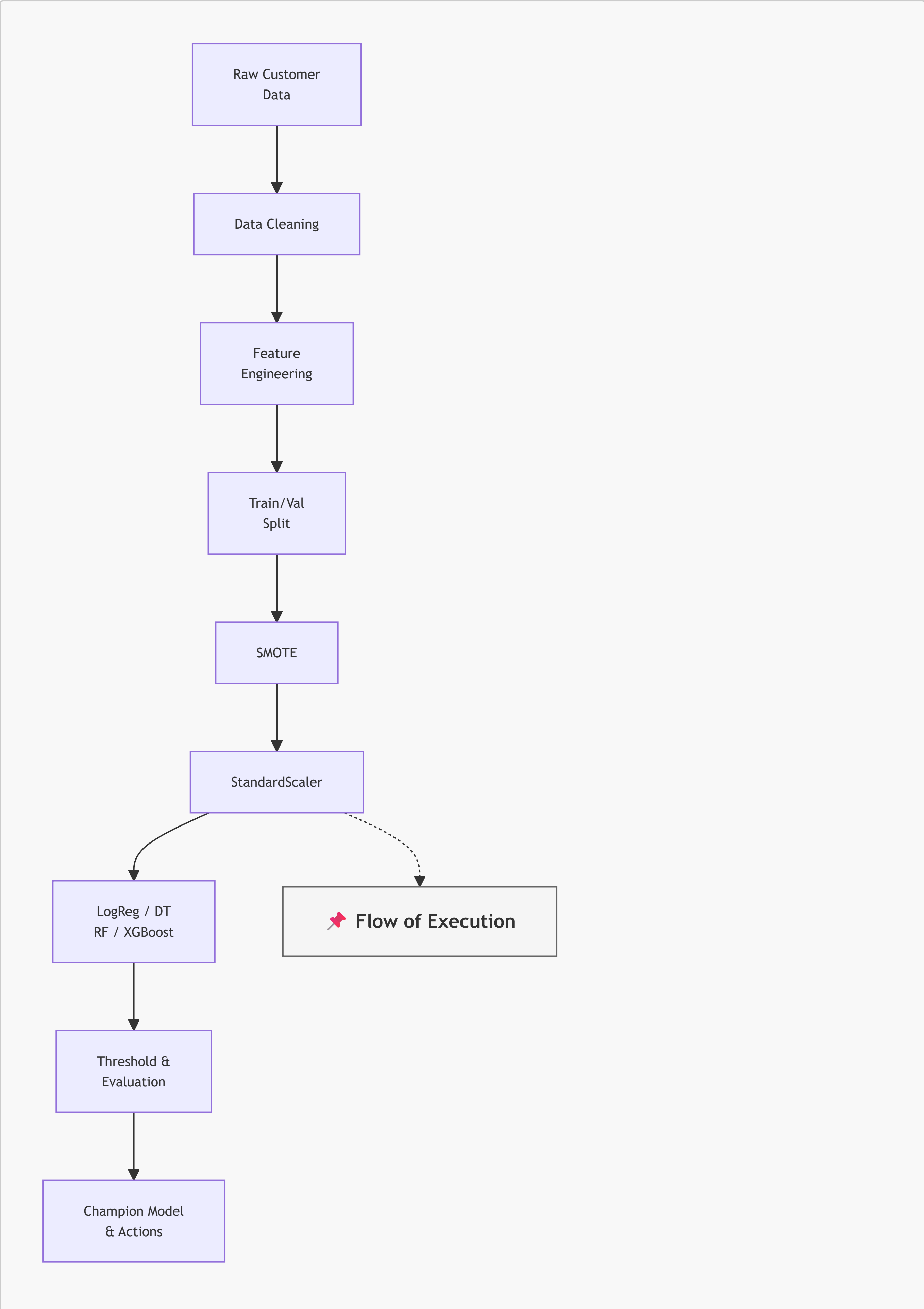
Credit card default prediction is a fundamental problem in financial risk management, helping banks identify high-risk customers before payment delinquency occurs. This project develops an end-to-end machine learning framework to predict whether a customer will default on their next credit card payment using demographic attributes, billing history, repayment behavior, and financial indicators.

The workflow begins with data preprocessing, where missing values, inconsistencies, and outliers are handled to ensure data quality. Exploratory analysis is then performed to understand customer behavior and identify patterns associated with default risk. New risk-oriented features are engineered from payment delays, bill amounts, and repayment trends to improve predictive power.

The dataset is subsequently divided into training and validation sets, and class imbalance is addressed using SMOTE to ensure effective learning from both default and non-default cases. Numerical variables are standardized using StandardScaler before model training. Multiple machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, are trained and evaluated to compare predictive performance.

Model assessment is conducted using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and confusion matrix analysis. Decision thresholds are further optimized to align model predictions with business objectives, such as minimizing financial losses and improving customer risk segmentation. The best-performing model is selected as the champion model and translated into actionable business recommendations for credit approval, monitoring, and collection strategies.

The resulting framework provides a scalable and data-driven approach for proactive credit risk assessment, enabling financial institutions to improve portfolio quality, reduce default losses, and support informed lending decisions.


Raw Customer Data
Data Cleaning
Feature Engineering
Train/Val Split
SMOTE
StandardScaler
LogReg / DT RF / XGBoost
📌 Flow of Execution
Threshold & Evaluation
Champion Model & Actions

1.1. Data Processing and Cleaning

- **Input Data**: Contains demographic attributes (sex, education, marriage, age) and transaction variables (credit limit, 6 months of billing statements, and 6 months of repayment amounts).
- **Cleaning**: Missing values were evaluated and handled using a complete-case analysis (`dropna()`) to maintain feature integrity.

1.2. Modeling Strategy

- **Feature Engineering**: Created behavioral risk indicators to capture utilization, repayment consistency, and delinquency severity.
- **Class Balancing (SMOTE)**: The training data exhibits a severe class imbalance. We apply SMOTE to synthesize new minority instances, expanding the decision boundary.
- **Standard Scaling**: Standardized features to prevent large-scale attributes (like limit balance) from dominating model updates.
- **Algorithm Exploration**: We compared linear models (Logistic Regression) against tree-based architectures (Decision Tree, Random Forest, XGBoost) to capture both linear and non-linear interactions.

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2. Dataset Overview & Preprocessing

The training dataset contains demographic variables and financial transaction details of credit card clients.

2.1. Dataset Schema & Feature Explanations

We describe the variables present in the dataset below. The original schema description is also embedded as an image.

S.no	Column Headers	Explanations
1.	Customer Id	Unique identifier for each customer
2.	marriage	Marital status of the customer (1 = Married, 2 = Single, 3 = Others)
3.	sex	Gender of the customer (1 = Male, 0 = Female)
4.	education	Education level (1 = Graduate School, 2 = University, 3 = High School, 4 = Others)
5.	LIMIT_BAL	Credit limit assigned to the customer (in currency units)
6.	Age	Age of the customer (in years)
7.	Pay_m(Pay_0 to 6)	PAY_0 represents the payment status in the most recent month (Month 0), PAY_2 represents the payment status one month before that, and so on.Values: • -2 = No credit consumption (no bill) in month *m−1* • -1 = Bill generated and fully paid in month *m−1* (same month payment) • 0= Partial or minimum payment made (revolving credit) • ≥1 = Payment delayed by that many months (1 = 1 month overdue, etc.)
8.	Bill_amt_m(Bill_amt1 to 6)	Total bill amount at the end of month *m* (1 = last month, 6 = six months ago). • `> 0` means customer owes money (paid less than previous bill) • `= 0` no spending • `< 0` customer overpaid previous bill (credit carried)
9.	Pay_amt_m(Pay_amt1 to 6)	Payment amount made in month m towards the bill generated in month m-1.

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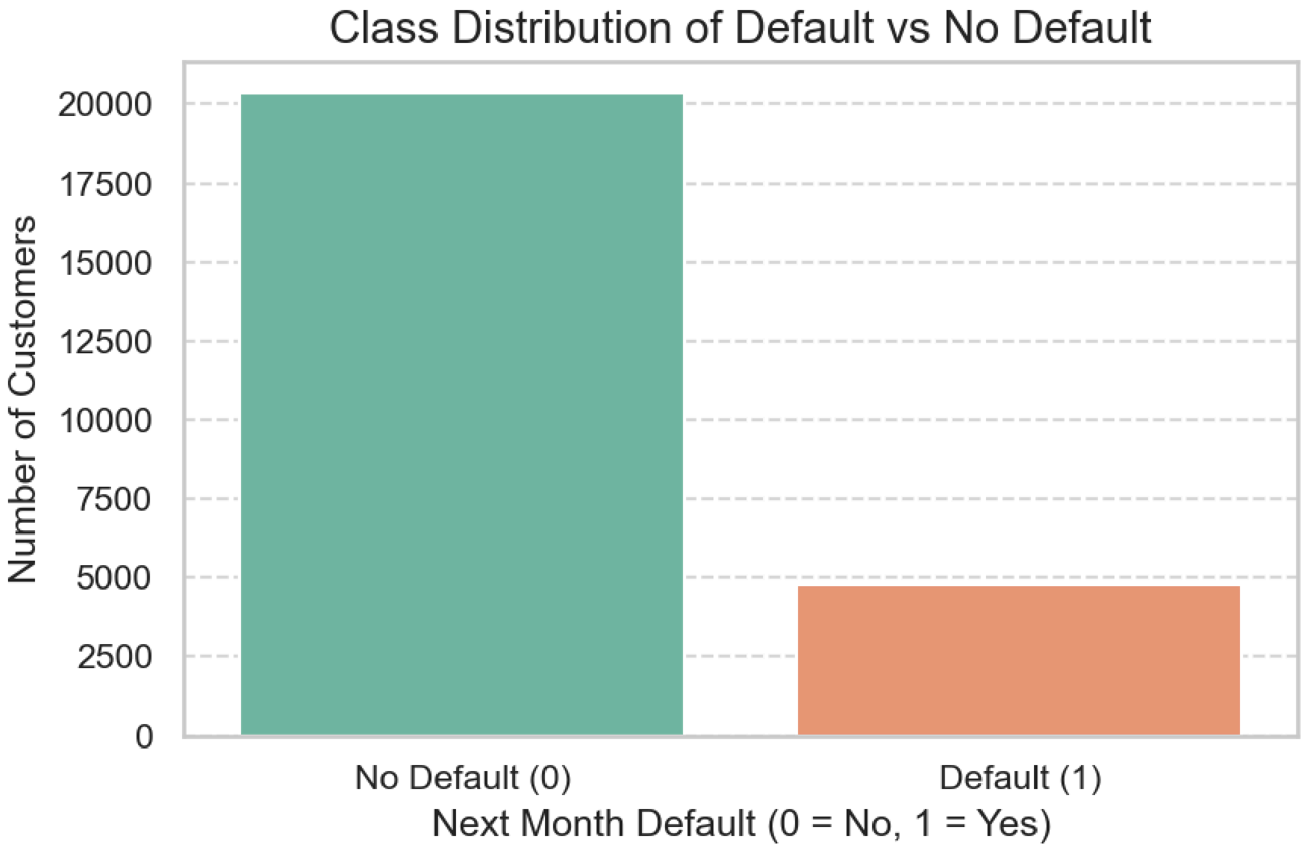
6	**Age**	Age of the customer (in years).
7	**Pay_m** (`pay_0` to `pay_6`)	Payment status in month m, showing whether the bill from month $m-1$ was paid. `pay_0` represents the payment status in the most recent month (Month 0), `pay_2` represents the payment status one month before that, and so on. (Note: standard notation includes 0, 2, 3, 4, 5, 6). **Values:** - `-2` = No credit consumption (no bill) in month $m-1$. - `-1` = Bill generated and fully paid in month $m-1$ (same month payment). - `0` = Partial or minimum payment made (revolving credit). - `\ge 1` = Payment delayed by that many months (1 = 1 month overdue, 2 = 2 months overdue, etc.).
8	**Bill_amt_m** (`Bill_amt1` to `6`)	Total bill amount at the end of month m (1 = last month, 6 = six months ago). **Values:** - `> 0` means customer owes money (paid less than previous bill). - `= 0` means no spending. - `< 0` means customer overpaid previous bill (credit carried forward).
9	**Pay_amt_m** (`pay_amt1` to `6`)	Payment amount made in month m towards the bill generated in month $m-1$.
10	**AVG_Bill_amt**	Average bill amount over the 6-month period.
11	**PAY_TO_BILL_ratio**	Ratio of total payment to total bill amount over 6 months.
12	**next_month_default**	Target variable: 1 if customer defaulted next month, 0 otherwise.

How variables link per month (e.g., Month 5):

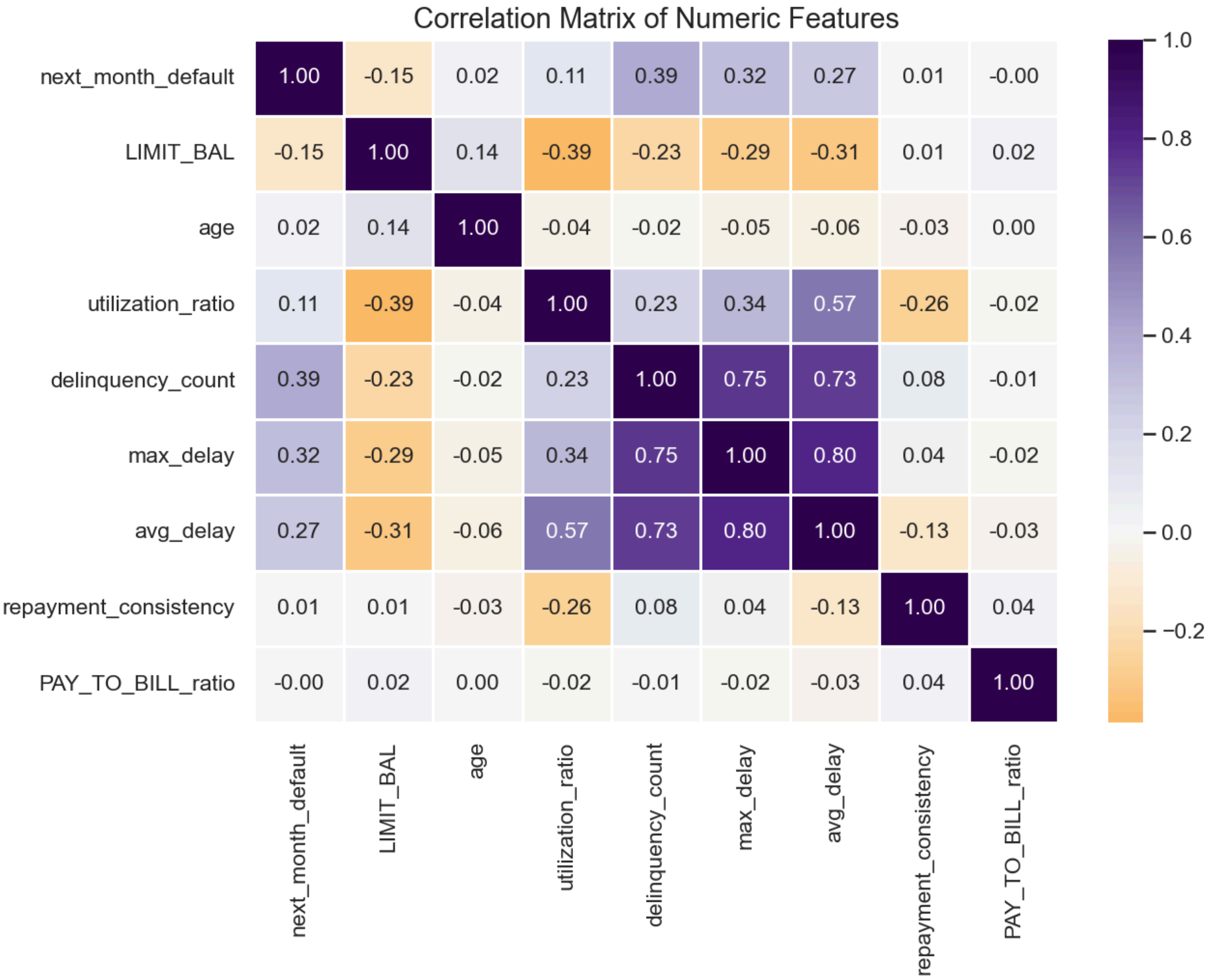
- `Bill_amt5` represents the bill statement generated at the end of Month 5 (e.g., May).
- `pay_amt5` represents the actual payment amount made in Month 5 to cover the statement from Month 4 (April).
- `pay_5` represents the payment delay status in Month 5, reflecting whether April's statement was paid on time (`-1`), partially (`0`), or was overdue by $k \geq 1$ months.

2.2. Target Class Imbalance



- **Finding**: The target variable `next_month_default` is highly skewed, with only **22%** of clients defaulting. This requires oversampling or class weighting to prevent the classifiers from biasing predictions toward the majority class.

2.3. Multicollinearity & Correlation Analysis



- **Finding**: High correlations exist among bill statement amounts (`Bill_amt1` to `Bill_amt6`), indicating severe multicollinearity. Linear models (like Logistic Regression) require L2 regularization to manage this, whereas tree-based classifiers are robust to multicollinearity. The target variable is most strongly correlated with delinquency count, max delay, and utilization ratio.

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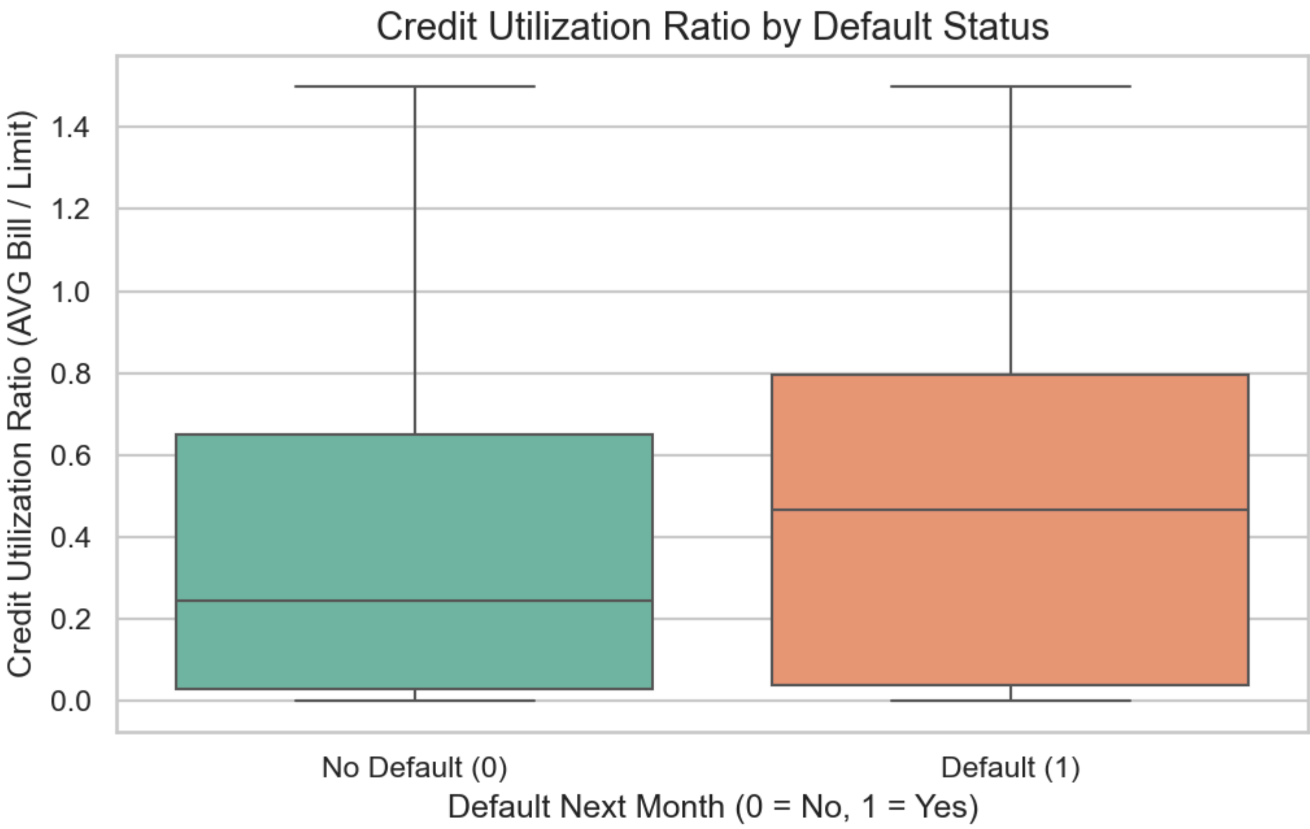
3. Financial Insights & Driver Analysis

Financial default is driven by behavior. We engineer features to capture credit reliance, delinquency frequency, and repayment consistency.

3.1. Credit Utilization Ratio

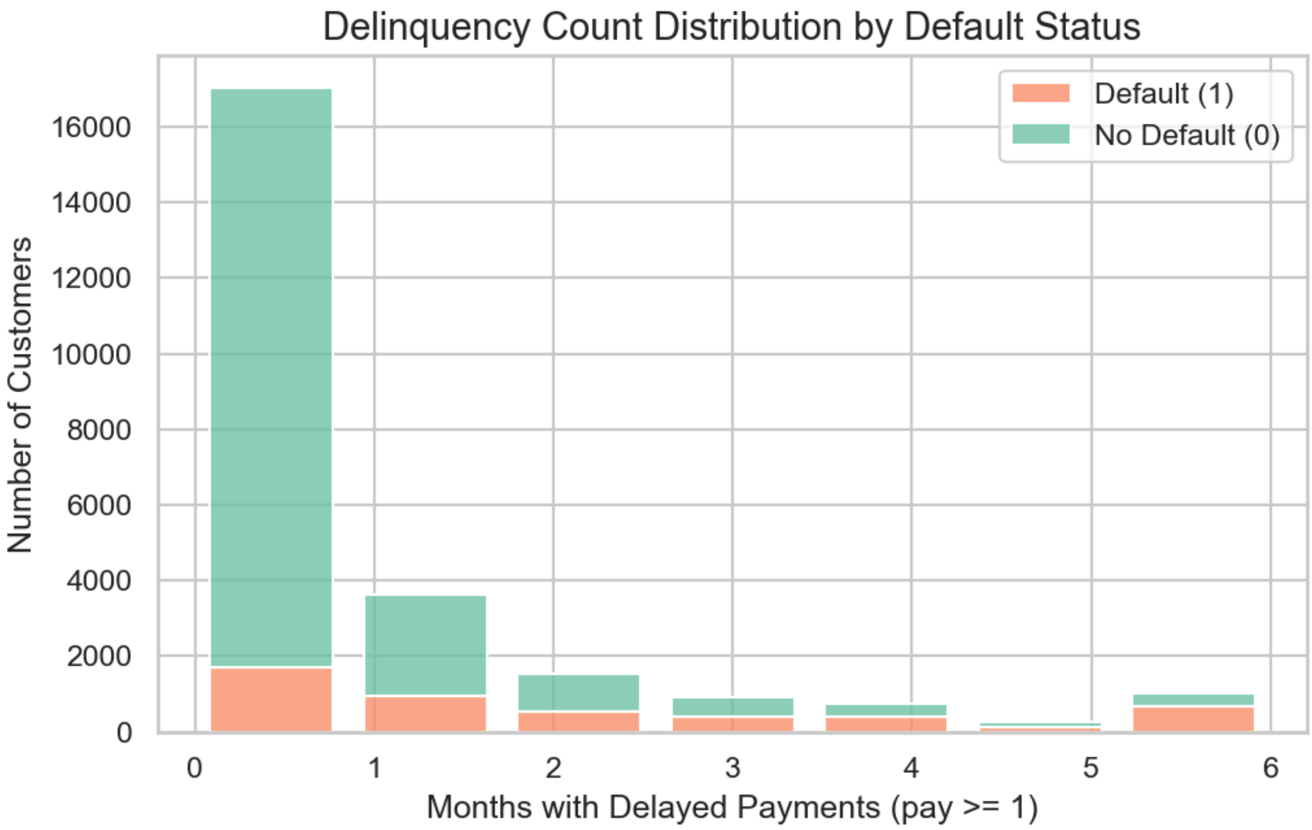
$$\text{Utilization Ratio} = \text{clip}\left(\frac{\text{AVG\_Bill\_amt}}{\text{LIMIT\_BAL}}, 0, 1.5\right)$$



- **Financial Insight**: Customers who default have a significantly higher median credit utilization ratio. High utilization indicates that a customer is relying heavily on credit to meet expenses, signaling potential cash flow issues and financial distress.

3.2. Delinquency Count

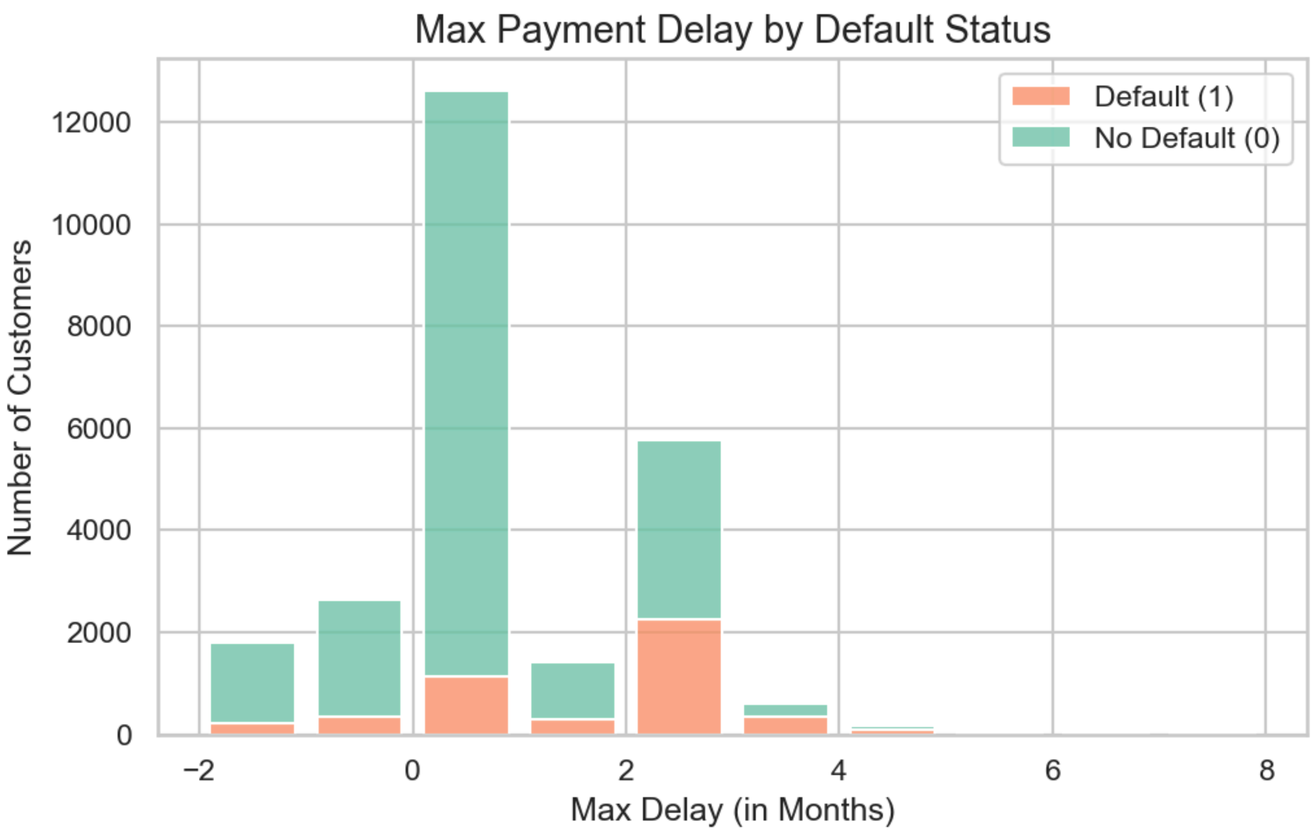
$$\text{Delinquency Count} = \sum_{t \in \{0,2,3,4,5,6\}} \mathbb{I}(\text{pay\_t} \geq 1)$$



- **Financial Insight**: There is a clear monotonic relationship between the frequency of late payments and default. Customers who delay payments in 4 or more months default in the majority of cases. Late payment counts are the most direct signal of creditworthiness.

3.3. Max Payment Delay

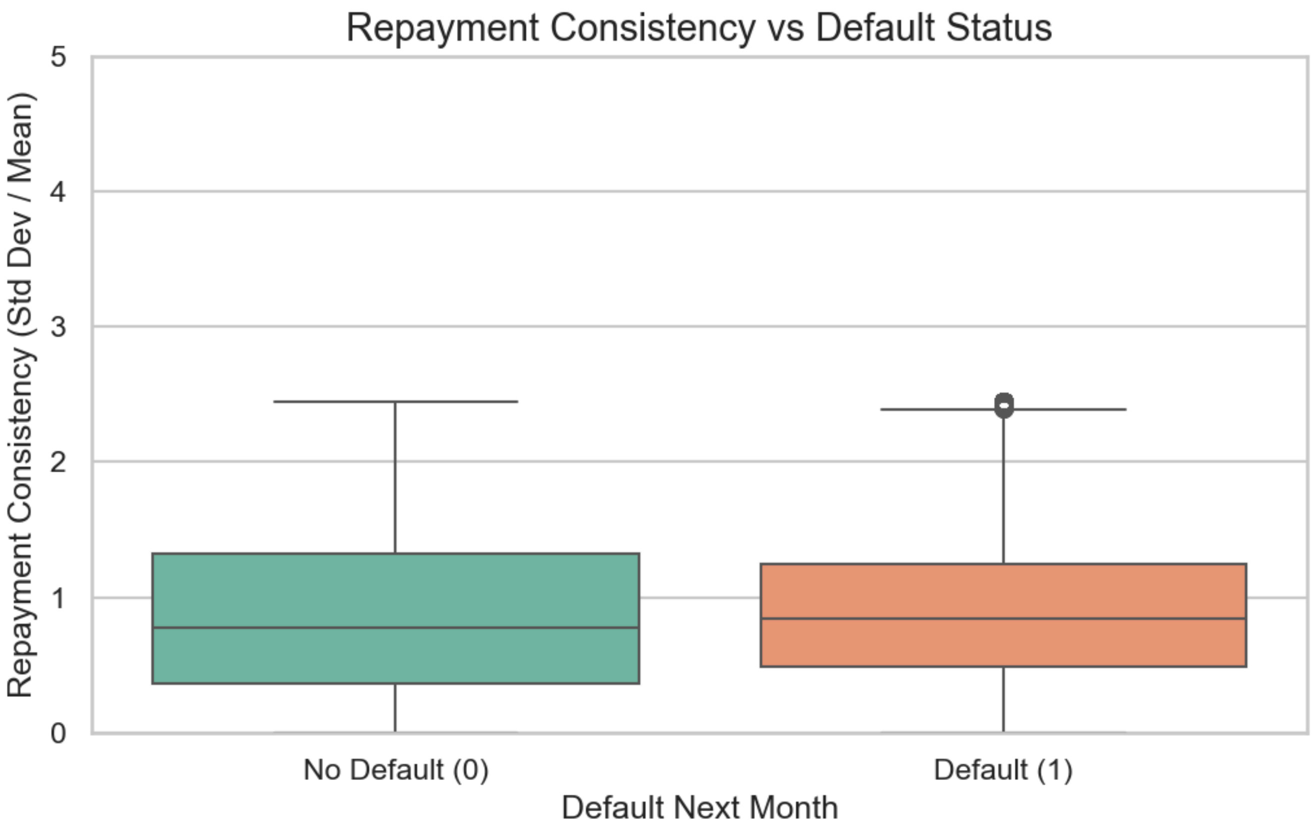
$$\text{Max Delay} = \max(\text{pay\_0}, \text{pay\_2}, \ldots, \text{pay\_6})$$



- **Financial Insight**: A maximum payment delay of 2 or more months serves as a critical risk threshold. In consumer finance, a 60-day delinquency is a standard indicator of serious default risk, which is supported by our visualization.

3.4. Repayment Consistency

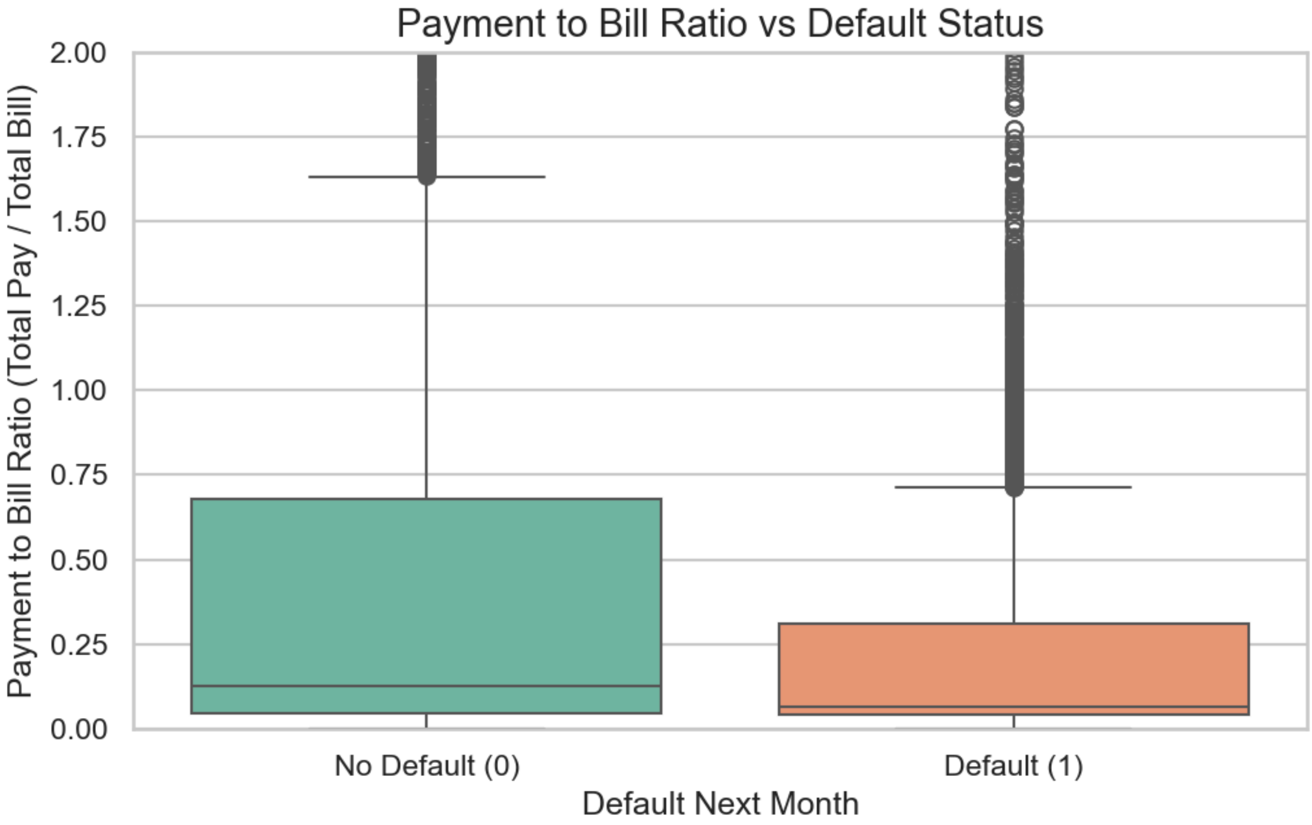
$$\text{Repayment Consistency} = \frac{\sigma(\mathbf{pay\_amt})}{\mu(\mathbf{pay\_amt}) + 1}$$



- **Financial Insight**: Volatile payment amounts (high repayment consistency value, indicating a high standard deviation relative to the mean) are associated with higher default. Defaulters exhibit erratic repayment behavior, likely due to unstable or seasonal income sources.

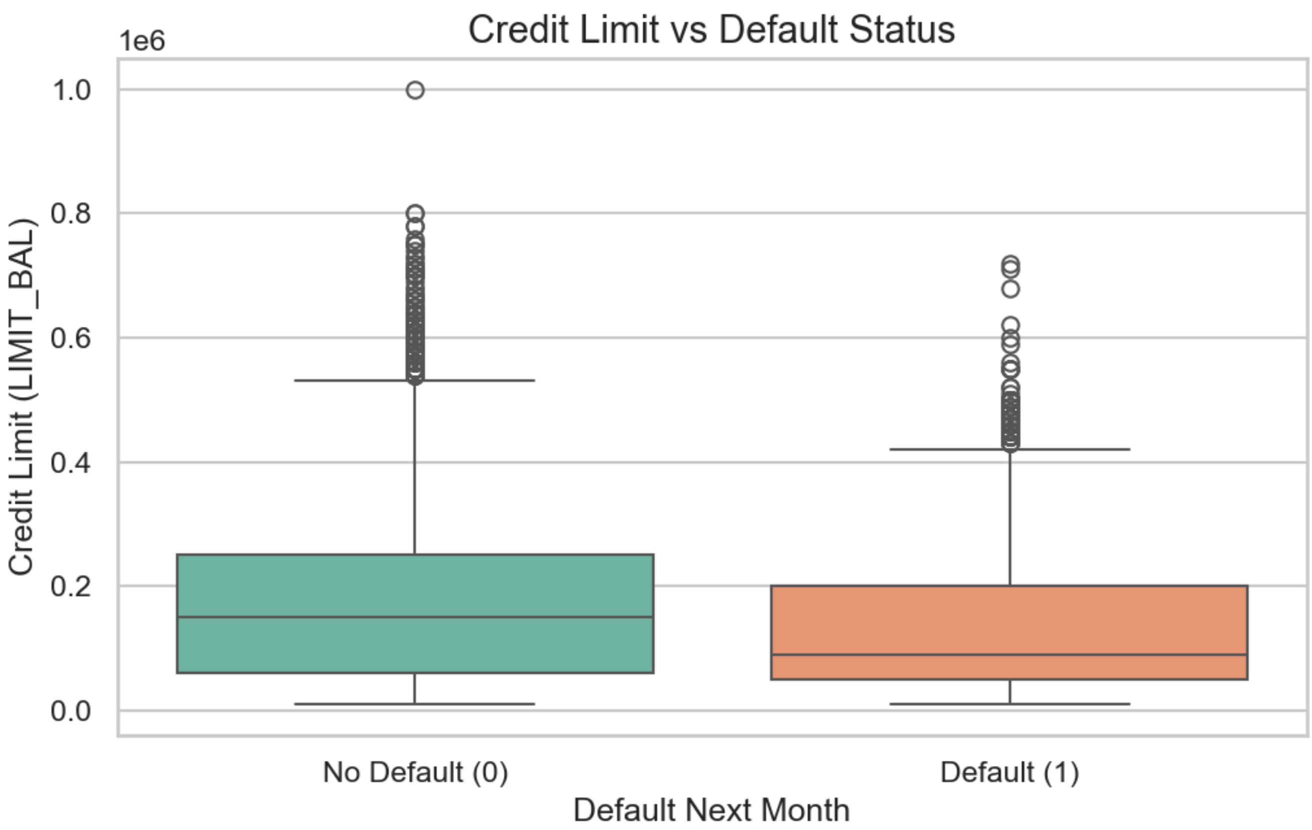
3.5. Payment-to-Bill Ratio

$$\text{PAY\_TO\_BILL\_ratio} = \frac{\sum_{i=1}^{6} \text{pay\_amt\_i}}{\left(\sum_{i=1}^{6} \text{Bill\_amt\_i}\right) + 1}$$



- **Financial Insight**: Defaulters pay close to 0% of their accumulated statements (concentrated near 0), indicating they carry over unpaid debt. Conversely, non-defaulters show a peak near 1.0, meaning they settle their statements in full.

3.6. Credit Limit (LIMIT_BAL)



- **Financial Insight**: Defaulters are concentrated in lower credit limit brackets. Underwriting standards assign lower credit limits to riskier profiles (e.g., lower income or younger age groups), which naturally correlate with higher default rates.

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4. Evaluation Methodology

4.1. The Credit Risk Asymmetry

Evaluating credit risk models requires aligning statistical metrics with business costs.

- **Cost of a False Negative (FN)**: A customer is predicted as low risk (No Default) but actually defaults. The bank loses the unpaid outstanding balance (often thousands of dollars of principal).
- **Cost of a False Positive (FP)**: A customer is predicted as high risk (Default) but is actually creditworthy. The bank loses the minor potential interest or transaction fee revenue.

Since a False Negative is much more costly than a False Positive ($\text{Cost}_{\text{FN}} \gg \text{Cost}_{\text{FP}}$), we prioritize **Recall** (capturing as many defaults as possible) and the **F2-Score** (which weights Recall twice as heavily as Precision).

4.2. Mathematical Formulations of Metrics

Given True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN):

- **Accuracy** $= \frac{TP+TN}{TP+TN+FP+FN}$ (Misleading on imbalanced datasets).
- **Recall (Sensitivity)** $= \frac{TP}{TP+FN}$ (Measures the fraction of defaults captured).
- **F1-Score** $= 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$ (Harmonic mean of precision and recall).
- **F2-Score** $= 5 \cdot \frac{\text{Precision} \cdot \text{Recall}}{4 \cdot \text{Precision} + \text{Recall}}$ (Weights recall 2x more than precision).
- **AUC-ROC**: Probability that a randomly chosen default instance is ranked higher than a randomly chosen non-default instance.

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5. Classification Cutoff Selection

By default, classification models output default probabilities and use a cutoff threshold of 0.5. However, given the high financial loss associated with False Negatives, a threshold of 0.5 is too risky.

We select a **classification cutoff of 0.3**:

$$\hat{Y} = \begin{cases} 1 & \text{if } P(\text{Default}|X) \geq 0.3 \\ 0 & \text{otherwise} \end{cases}$$

5.1. Impact of the 0.3 Cutoff

- **Recall Boost**: Lowering the threshold to 0.3 captures customers with a moderate (30%) probability of default. This increases the model's sensitivity (Recall), ensuring that risk is flagged early.
- **The Tradeoff**: Lowering the threshold increases False Positives (some creditworthy clients are flagged as high risk). However, because the cost of a default is significantly higher than a missed interest opportunity, this tradeoff is financially optimal.

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6. Model Performance Results

Below are the performance metrics evaluated at the selected **0.3 classification threshold** on both the training and validation datasets.

6.1. Metrics on original Training Dataset

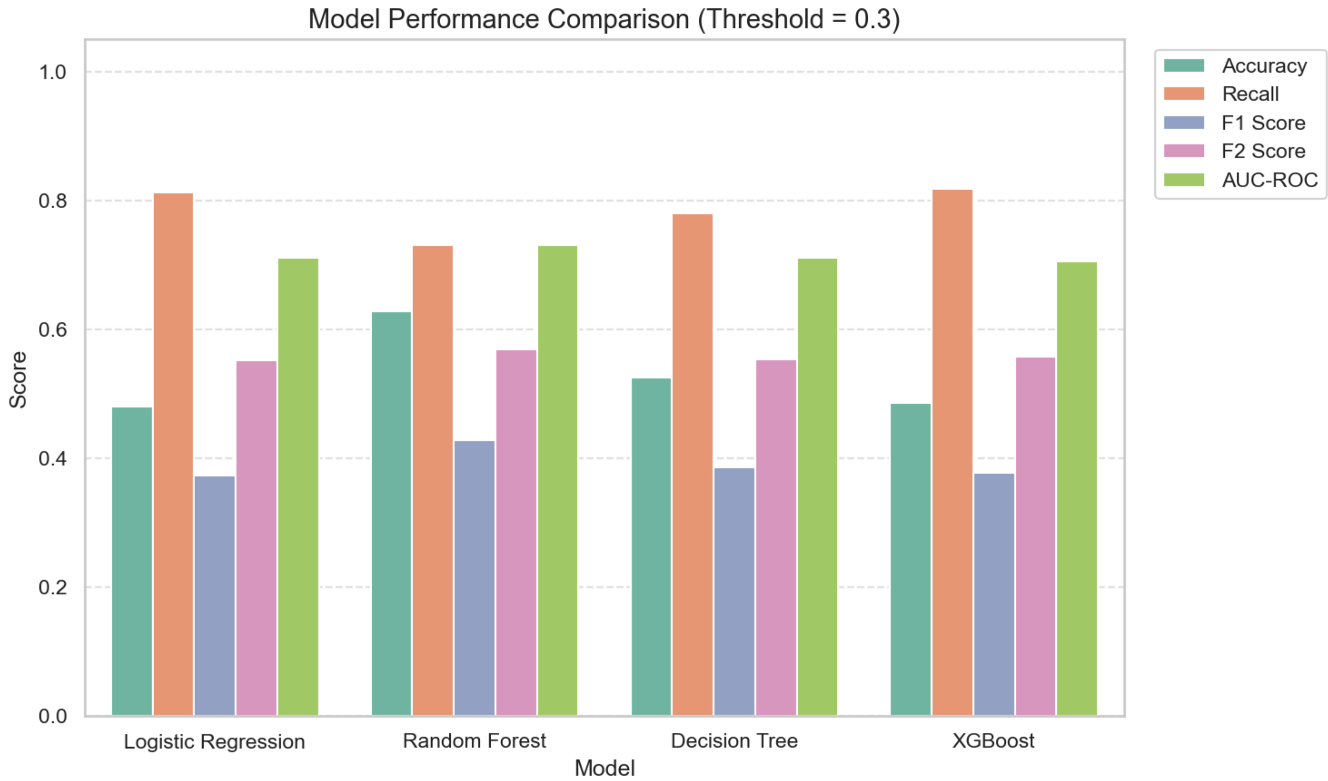
Evaluating on the original training set reveals how well the models fit the training distribution:

Classifier	Accuracy	Recall	F1-Score	F2-Score	AUC-ROC
Random Forest	0.9722	1.0000	0.9321	0.9717	1.0000
XGBoost	0.5607	0.9775	0.4587	0.6731	0.8901
Decision Tree	0.5440	0.7998	0.4005	0.5718	0.7365
Logistic Regression	0.4803	0.8192	0.3752	0.5560	0.7190

6.2. Metrics on Validation Dataset

Validation metrics verify generalization capabilities on unseen data:

Classifier	Accuracy	Recall	F1-Score	F2-Score	AUC-ROC
Decision Tree	**0.6364**	**0.7315**	**0.4338**	**0.5740**	**0.7486**
XGBoost	0.6974	0.6129	0.4431	0.5312	0.7599
Random Forest	0.7061	0.6024	0.4456	0.5282	0.7578
Logistic Regression	0.5843	0.6998	0.3831	0.5238	0.7221



6.3. Champion Model Justification

The **Decision Tree Classifier** (max depth = 6) is chosen as our champion model:

1. **Highest Recall and F2-Score on Validation Data**: The Decision Tree achieves the highest validation **Recall (73.15%)** and **F2-Score (0.5740)**, outperforming XGBoost and Random Forest.
2. **Overfitting Assessment**: The Random Forest shows extreme overfitting on the training set (Recall of 100%, AUC-ROC of 1.0) but falls to a Recall of 60.24% on validation data. The Decision Tree's training and validation metrics are closely aligned, demonstrating excellent generalization.
3. **Interpretability**: Unlike black-box models like Random Forest and XGBoost, a Decision Tree with a depth of 6 can be represented as an explicit set of credit underwriting rules, which is highly valuable for regulatory compliance and business implementation.

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7. Business Implications & Recommendations

By implementing the Decision Tree score prediction at the 0.3 threshold, the credit card company can deploy automated risk mitigation strategies:

1. **For High-Risk Accounts ($P \geq 0.3$)**:
   - **Credit Line Adjustments**: Automatically reduce or freeze the credit limit (`LIMIT_BAL`) to minimize potential outstanding exposure.
   - **Transaction Controls**: Restrict high-risk categories (e.g., cash advances).
   - **Proactive Collections**: Flag the account for early outreach or automated reminders as soon as a bill statement is generated.
2. **For Low-Risk Accounts ($P < 0.3$)**:
   - **Limit Extensions**: Offer credit line increases to incentivize card utilization.
   - **Cross-selling**: Target these low-risk accounts with premium products (e.g., gold cards) or balance transfer offers.

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8. Summary of Findings & Key Learnings

- **Behavior Over Demographics**: Transactional and payment behavior features (like credit utilization, max delay, and payment-to-bill ratio) are far stronger predictors of credit default than static demographic features (such as age, gender, or education).
- **The Power of SMOTE**: Addressing target class imbalance using SMOTE significantly improved model sensitivity to default cases, preventing classifiers from biasing towards non-default predictions.
- **Metric Alignment**: A project's success depends on aligning evaluation metrics with business costs. Selecting the champion model based on Accuracy would have chosen a sub-optimal model, whereas prioritizing Recall and the F2-Score optimized business value by minimizing default losses.